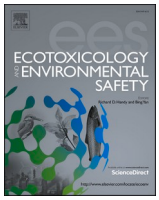
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Exposome-wide association study of environmental toxicants and periodontitis among United States adults

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ABSTRACT

Objectives: To systematically evaluate associations between environmental toxicants and periodontitis and to quantify the mediating role of systemic inflammatory biomarkers.
Methods: We extracted data from National Health and Nutrition Examination Survey (NHANES) 2013–2014. The study covers 73 environmental toxicants spanning 12 categories. Exposome-wide association study (ExWAS) was used to identify individual toxicant associations, and adaptive elastic net regularization was employed to develop an environmental risk score (ERS). Mediation analysis quantified the contribution of inflammatory biomarkers to observed associations
Results: Among 1210 participants, 49.4 % were male and 50.6 % were female. The ExWAS identified 29 toxicants across 9 categories significantly associated with periodontitis. Notably, detectable monohydroxybutenyl mercapturic acid (MHBMA2) levels were associated with a 2.798-fold higher periodontitis odds (95 % CI: 1.799–4.352) versus undetectable levels. Stratified analyses revealed that associations were stronger among males and younger adults (< 60 years). The ERS, derived from 17 prioritized toxicants via elastic net regression, demonstrated a significant positive dose-response relationship with periodontitis risk, where each IQR increase was associated with a 59.7 % higher disease risk (OR = 1.597). Mediation analysis indicated that white blood cell count (11.5 %), neutrophils (10.2 %), monocytes (5.1 %), systemic immune-inflammation index (6.3 %), neutrophil-to-lymphocyte ratio (5.7 %), and systemic inflammation response index (7.4 %) significantly mediated toxicant-periodontitis associations.
Conclusions: This nationally representative exposome-wide investigation identified significant associations between multiple environmental toxicants and periodontitis, with inflammatory pathways partially mediating these relationships. These findings establish potential environmental etiological factors for periodontitis and highlight specific toxicants as targets for exposure reduction strategies to improve periodontal health at the population level.

1. Introduction

Periodontitis is a chronic inflammatory disease caused by microorganisms present in dental plaque, characterized by the destruction of connective tissue attachment and alveolar bone loss (Slots, 2017). The disease represents a significant global health burden(Yan et al., 2025c), with advanced stages potentially leading to tooth mobility and loss, significantly impairing masticatory function and quality of life (Yin

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et al., 2022). Furthermore, growing evidence demonstrates a bidirectional relationship with systemic conditions such as diabetes, cardiovascular diseases, and neurological disorders(Herrera et al., 2023; Liccardo et al., 2020; Yan et al., 2025a). According to the 2009–2014 National Health and Nutrition Examination Survey (NHANES), approximately 42 % of U.S. adults suffer from periodontitis, with severe cases accounting for 7.8 %(Eke et al., 2018). Given this substantial prevalence and its systemic implications, periodontitis has emerged as a significant public health concern, necessitating effective preventive strategies to understand and mitigate its risk factors.

While traditional risk factors such as smoking, poor oral hygiene, dietary factors, and genetic predisposition have been well-established (Amarasena et al., 2002; Chen et al., 2024; Wu et al., 2023; Yan et al., 2025b), mounting evidence suggests that environmental pollutant exposure represents an emerging and underappreciated contributor to periodontal disease pathogenesis. Studies indicate that exposure to hundreds of natural and synthetic toxicants, including heavy metals (lead, mercury, cadmium, arsenic, chromium, etc.)(Yang et al., 2024a, 2024b), ethylene oxide(Miao et al., 2024), brominated flame retardants, (Pan et al., 2024) polycyclic aromatic hydrocarbons(Wu et al., 2024), and volatile organic compounds(Dong et al., 2024), are associated with periodontitis. However, existing studies have predominantly adopted a single-exposure approach, examining individual chemicals or isolated chemical classes. This methodological limitation fails to capture the real-world complexity of simultaneous exposure to multiple environmental toxicants, potentially leading to underestimated effect sizes and omitted variable bias.

The exposome concept, introduced in 2005, offers a novel approach to address this methodological limitation. This framework systematically evaluates the health effects of all environmental exposures (i.e., non-genetic factors), providing a scientific basis for understanding the biological consequences of mixed exposures while minimizing selective reporting and confounding biases(Vermeulen et al., 2020; Wild, 2005). Despite the conceptual appeal and analytical advantages of the exposome approach, no comprehensive exposome-wide association study (ExWAS) has specifically investigated the relationship between complex environmental mixtures and periodontitis in a representative population sample.

To address this critical knowledge gap, we conducted an ExWAS utilizing data from the 2013–2014 NHANES to comprehensively examine associations between environmental toxicant exposure and periodontitis in United States adults. Our study aims to systematically identify environmental toxicants associated with periodontitis risk, assess the combined effects of multiple exposures, and provide insights into the complex interplay between environmental factors and periodontal health that could inform future prevention strategies and public health policies.

2. Methods

2.1. Study population

This study utilized data from the 2013–2014 cycle of the NHANES, a stratified, multistage probability sampling survey designed to generate nationally representative estimates of the noninstitutionalized US population. NHANES is conducted by the Centers for Disease Control and Prevention's National Center for Health Statistics (NCHS), with study protocols approved by the NCHS Institutional Ethics Review Board and written informed consent obtained from all participants.

The analytical cohort comprised survey respondents aged 20 years or older who had complete periodontitis examination data and environmental toxicant measurements. Individuals with missing periodontitis data or missing values for environmental toxins exceeding one-third of the analytes were excluded, resulting in a final sample of 1210 participants (Fig. 1). Data are publicly available in NHANES and can be accessed at: https://www.cdc.gov/nchs/nhanes/index.htm.

2.2. Periodontitis assessment

Periodontitis status was determined using clinical examination data from NHANES, specifically derived from the periodontal file ("OHXPER. Doc"). Classification followed the surveillance case definitions jointly established by the Centers for Disease Control and Prevention and the American Academy of Periodontology (CDC/AAP)(Du et al., 2023; Eke et al., 2015), based on thresholds of clinical attachment loss (CAL) and periodontal probing depth (PPD). Severe periodontitis was defined as CAL $\geq$ 6 mm in $\geq$ 2 proximal sites (affecting different teeth) and PPD $\geq$ 5 mm in $\geq$ 1 proximal site. Moderate periodontitis was defined as $\geq$ 2 proximal sites (affecting different teeth) with CAL $\geq$ 4 mm or PPD $\geq$ 5 mm. Mild periodontitis was defined as CAL $\geq$ 3 mm in $\geq$ 2 proximal

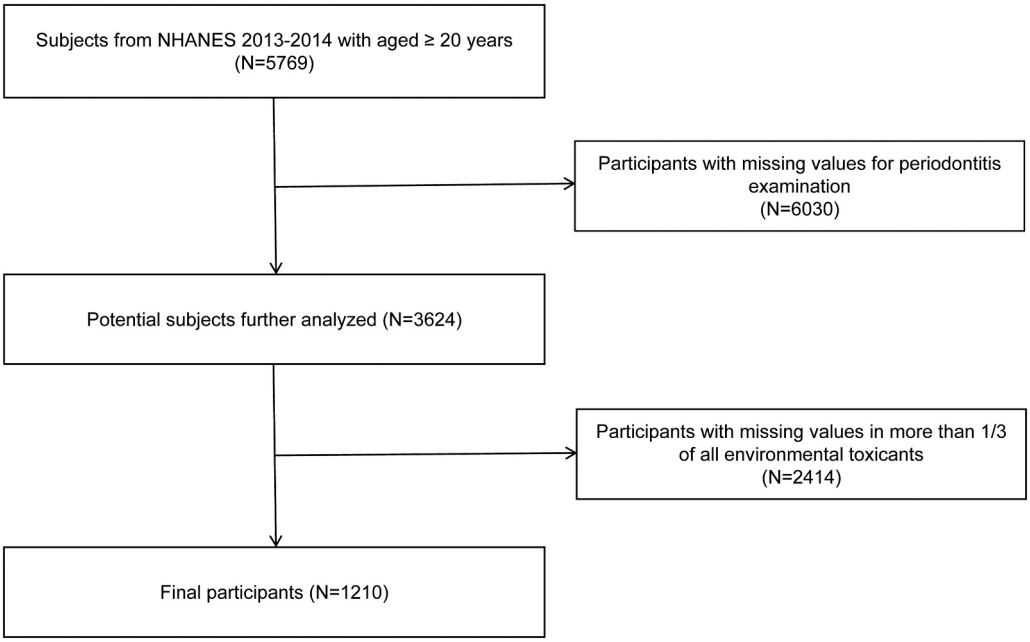


Fig. 1. Flow chart of the study population in this study. Abbreviation: NHANES, National Health and Nutrition Examination Survey.

sites (different teeth), plus PPD ≥ 4 mm in ≥ 2 proximal sites (different teeth) or PPD ≥ 5 mm in ≥ 1 site.

In this study, participants were categorized as having periodontitis (mild, moderate, or severe) or non-periodontitis.

2.3. Exposure assessment

Environmental toxicant exposure was assessed through biochemical analysis of blood and urine specimens collected during the NHANES examination. Initially, 89 environmental toxicants across 13 chemical categories were evaluated (Supplementary Table 1). Analytical methodologies included: quantification of hemoglobin adducts for reactive organic compounds (e.g., acrylamide, glycidamide, ethylene oxide, formaldehyde), expressed as adduct quantity per milligram of hemoglobin; and determination of urinary concentrations for inorganic elements and metallic toxicants, adjusted for urinary creatinine concentration to account for dilution effects. Following standardized procedures for handling values below the limit of detection and missing data (Methods in Supplementary material), 73 toxicants across 12 chemical categories were retained for primary analysis.

2.4. Covariates

Demographic, socioeconomic, behavioral, and clinical covariates were selected based on established associations with periodontitis and potential confounding relationships. These included: age, gender, race, education level, marital status, poverty-income ratio, BMI, alcohol consumption, the time since last dental visit, mouthwash use, flossing frequency, and estimated glomerular filtration rate (eGFR). A directed acyclic graph (DAG) was developed to assess potential confounding structures (Supplementary Figure 1). A more detailed explanation of covariates is in the Methods section of the supplementary material.

2.5. Statistical analyses

All analyses accounted for the complex survey design of NHANES (2013–2014 cycle), including sample weights (wtsa2yr), stratification variables, and primary sampling units (PSUs) to ensure nationally representative estimates, unless otherwise specified for methodological reasons. This applied to descriptive statistics, restricted cubic spline (RCS) analyses, subgroup analyses, and mediation analysis.

An exposome-wide association study (ExWAS) was conducted using the "rexposome" package to systematically evaluate associations between individual environmental toxicants and periodontitis (Hernandez-Ferrer et al., 2019). All associations were assessed using two-sided *p*-values, with multiple testing correction performed via the Benjamini-Hochberg procedure to control the false discovery rate (FDR) (Tzoulaki et al., 2012). Within the ExWAS framework, which involves fitting independent regression models for each exposure, we reported unweighted results to maintain methodological consistency with prior investigations and to avoid potential over-adjustment bias(Guo et al., 2024; James-Todd et al., 2012; Lewis et al., 2015).

The Environmental Risk Score (ERS) was constructed using an elastic net regression framework implemented via the "gcdnet" package. We performed comprehensive parameter tuning by systematically evaluating 11 α values (ranging from 0 to 1) and 50 λ values (from 10^{3} to 10^{-2}). The optimal parameter combination was identified through 5-fold cross-validation. To evaluate the robustness of variable selection, we conducted a bootstrap analysis with 1000 resampling iterations and calculated the inclusion frequency for each variable. Participants were dichotomized into high/low ERS groups based on median value. The association between ERS and periodontitis was evaluated using weighted logistic regression in 2 models: Model 1 was unadjusted, and Model 2 adjusted for all aforementioned covariates.

To evaluate potential nonlinearity, we employed restricted cubic splines (RCS) with 3 knots to model the association between the ERS and periodontitis. As a complementary mixture analysis, we conducted Weighted Quantile Sum (WQS) regression using the gWQS package, setting the number of quantiles to 4 and using 1000 bootstrap iterations.

Mediation analysis was explicitly framed as exploratory and was conducted to generate hypotheses rather than establish causal pathways. We used the "mediation" package in R with 1000 bootstrap iterations. A formal sensitivity analysis was performed using the medsens function to assess the robustness of findings to potential unmeasured confounding.

In all association analyses presented in the main text, we report odds ratios (ORs) per interquartile range (IQR) increase in environmental exposures to facilitate clinical interpretation. Effect estimates based on original measurement units are provided in the Supplementary Material. E-values were calculated to quantify the potential impact of unmeasured confounding required to explain away the observed associations.

All statistical analyses were performed using R software (version 4.5.0). Statistical significance was defined as two-tailed $P < 0.05$, unless otherwise specified for multiple comparisons using false discovery rate correction. More details are described in the Methods in Supplementary material.

3. Results

3.1. Characteristics of the study population

Table 1 presents general characteristics of 1210 NHANES 2013–2014 participants (49.4 % male, 50.6 % female). Significant differences ($P < 0.05$) between the non-periodontitis and periodontitis groups were observed for sex, age, race/ethnicity, education level, marital status, poverty-income ratio, dental floss use, time since last dental visit, and BMI ($P < 0.05$). Participants with periodontitis were more likely to be male, aged ≥ 60 years, have higher proportions of overweight/obesity, and report significantly lower frequencies of dental floss use and dental visits compared to non-periodontitis individuals. No significant intergroup differences ($P > 0.05$) were found in eGFR, alcohol consumption, or mouthwash use.

3.2. Data preprocessing and correlation structure of environmental toxicants

Of the 89 environmental toxicants initially evaluated, 73 met inclusion criteria based on data sufficiency. 16 toxicants underwent dichotomization based on detection limits, while the remaining 57 were log-transformed (natural logarithm) to approximate normal distribution (Supplementary Table 1). As shown in Supplementary Figure 2, varying degrees of correlation existed among the 73 environmental toxicants investigated.

3.3. Exposome-wide association Study (ExWAS) Findings

In the fully adjusted ExWAS model, which accounted for multiple testing using the Benjamini-Hochberg false discovery rate (FDR) procedure, we identified 29 toxicants significantly associated with periodontitis risk (FDR-corrected *q*-value < 0.05). These significant toxicants spanned nine distinct chemical categories: acrylamide, aromatic amines (n = 3), nicotine metabolites (n = 3), ethylene oxide, heterocyclic aromatic amines (n = 2), metals (n = 3), polycyclic aromatic hydrocarbons (n = 6), perchlorate, nitrate and thiocyanate compounds (n = 1), and volatile organic compound (VOC) metabolites (n = 9). A sensitivity analysis implementing family-wise FDR control within each chemical class corroborated the robustness of these findings (Supplementary Table 2). Notably, individuals with detectable levels of the VOC metabolite MHBMA2 exhibited a 2.798-fold increased risk of periodontitis (95 % CI: 1.799–4.352) compared to those with undetectable levels. Furthermore, each IQR increase in 2-Aminonaphthalene was associated with an 85.2 % higher probability of periodontitis (OR:

Table 1
Demographic characteristics of the study population in NHANES 2013-2014.

Variables	Total (n = 1210)	Nonperiodontitis (n = 677)	Periodontitis (n = 533)	*P*
eGFR (mL/min/1.73 m^2), median (Q25 %, Q75 %)	90.3 (77.0,104.1)	89.7 (76.9103.8)	91.2 (77.1104.7)	0.945
Sex, n (%)				**0.003**
Female	606 (50.6)	397 (55.8)	209 (41.3)	
Male	604 (49.4)	280 (44.2)	324 (58.7)	
Age (years), n (%)				**< 0.001**
<60	837 (72.5)	522 (77.1)	315 (64.4)	
≥ 60	373 (27.5)	155 (22.9)	218 (35.6)	
Ethnicity, n (%)				**0.004**
Mexican American	176 (9.2)	88 (7.9)	88 (11.6)	
Non-Hispanic Black	238 (10.9)	107 (8.6)	131 (15.0)	
Non-Hispanic White	523 (67.3)	319 (70.8)	204 (61.0)	
Other	273 (12.6)	163 (12.7)	110 (12.4)	
Education, n (%)				**< 0.001**
High school	254 (20.9)	122 (17.0)	132 (27.8)	
Less than high school	246 (13.2)	89 (8.5)	157 (21.7)	
Some college or above	710 (65.9)	466 (74.5)	244 (50.5)	
Marital, n (%)				**0.030**
Married	700 (62.3)	413 (65.8)	287 (56.0)	
Never married	160 (12.0)	84 (10.1)	76 (15.3)	
Unmarried but have/had partner	350 (25.7)	180 (24.1)	170 (28.7)	
Poverty, n (%)				**< 0.001**
<1.3	347 (19.4)	140 (13.6)	207 (33.0)	
> 3.5	403 (43.4)	301 (57.0)	102 (26.3)	
1.3–3.5	377 (31.7)	191 (29.4)	186 (40.8)	
Alcohol use, n (%)				0.568
No	166 (10.4)	95 (11.5)	71 (10.0)	
Yes	970 (84.2)	545 (88.5)	425 (90.0)	
Mouthwash use, n (%)				0.950
No	509 (43.6)	284 (43.7)	225 (43.4)	
Yes	701 (56.4)	393 (56.3)	308 (56.6)	
Dental floss use, n (%)				**0.030**
No	350 (25.0)	155 (21.5)	195 (31.1)	
Yes	860 (75.0)	522 (78.5)	338 (68.9)	
Last time visit a dentist, n (%)				**< 0.001**
1 year or less	730 (65.1)	477 (75.0)	253 (47.4)	
More than 1 year	480 (34.9)	200 (25.0)	280 (52.6)	
BMI (kg/m^2), n (%)				**0.029**
Normal	337 (27.0)	201 (30.4)	136 (21.3)	
Obese	489 (38.8)	271 (36.0)	218 (44.2)	
Overweight	378 (33.8)	204 (33.6)	174 (34.5)	

Abbreviations: NHANES, National Health and Nutrition Examination Survey; BMI, body mass index; eGFR, estimated glomerular filtration rate.

1.852, 95 % CI: 1.413–2.426), and each IQR increase in Total Cotinine was associated with an 84.9 % increased risk (OR: 1.849, 95 % CI: 1.477–2.315). The detailed results of the ExWAS are presented in Table 2 and visualized in the volcano plot in Fig. 2.

3.4. Stratified analyses by age and sex

The analysis was stratified by age and sex. The findings suggest that more types of environmental toxicants were associated with periodontitis among men (25 toxicants) and individuals younger than 60 years (32 toxicants) compared with women and older adults, respectively, particularly for VOC metabolites and Polycyclic Aromatic Hydrocarbons (Fig. 3 and Supplementary Table 3–4).

3.5. Construction and validation of the environmental risk score (ERS)

An ERS was constructed using 17 priority pollutants identified through adaptive elastic-net regularization. Bootstrap resampling (1000 iterations) demonstrated high variable selection stability for a core set of toxicants, including o-Toluidine, 2-Naphthol, 2-Aminonaphthalene, Lead, and TTCA, all with inclusion frequencies exceeding 80 % (Supplementary Table 5). Both continuous and categorical ERS measures demonstrated significant positive associations with periodontitis ($P < 0.05$). After adjusting for potential confounders, robust associations persisted, with each interquartile range (IQR) increase in the continuous ERS associated with a 59.7 % higher risk of periodontitis (OR = 1.597; 95 % CI: 1.418–1.800), while participants in the high ERS group had a 2.552-fold increased risk (95 % CI: 1.894–3.440) compared to those in the low ERS group. Restricted cubic spline analysis revealed no evidence of nonlinearity (P for nonlinear=0.348); however, periodontitis risk demonstrated a consistent dose-response relationship with increasing ERS levels. Furthermore, higher ERS values correlated with decreased tooth count and increased probing pocket depth and clinical attachment loss (Fig. 4).

Subgroup analyses demonstrated consistent positive associations between ERS and periodontitis risk across most demographic and behavioral strata after confounder adjustment ($P < 0.05$), with exceptions noted among high school graduates and non-drinkers (Supplementary Figure 3). No significant effect modification was detected in stratified analyses.

3.6. Exploratory mediation analysis by immunoinflammatory biomarkers

Mediation analyses evaluated the role of immunoinflammatory biomarkers in the ERS-periodontitis association. All six assessed mediators demonstrated statistically significant indirect effects: white blood cell count (WBC), neutrophil count (NEUT), monocyte count (MONO), systemic immune-inflammation index (SII), neutrophil-to-lymphocyte ratio (NLR), and systemic inflammation response index (SIRI). The respective mediation proportions were 11.494 %, 10.188 %, 5.050 %, 6.317 %, 5.710 %, and 7.351 % (Fig. 5 and Supplementary Table 6).

To assess the robustness of these findings against potential unmeasured confounding, formal sensitivity analyses were conducted. The results indicated that the observed mediation effects might be

Table 2
Exposome-wide association study of 73 environmental toxicants exposure with periodontitis among 1210 US adults.

Exposure family by toxicant exposure	Processing	Median (IQR)	Model 1			Model 2		
			OR (95 %CI)	*p-values*	*q-values*	OR (95 %CI)	*p-values*	*q-values*
Acrylamide & Glycidamide								
Acrylamide	log (ln)	-0.633 (-1.235, 0.070)	1.122 (0.945, 1.331)	0.188	0.240	1.307 (1.049, 1.629)	0.016	0.042
Glycidamide	log (ln)	-0.797 (-1.373, −0.154)	1.021 (0.865, 1.206)	0.804	0.849	1.209 (0.975, 1.499)	0.083	0.160
Aromatic Amines								
1-Aminonaphthalene	2 categories	NA	1.778 (1.370, 2.307)	< 0.001	< 0.001	1.336 (0.961, 1.856)	0.084	0.160
2-Aminonaphthalene	log (ln)	1.543 (0.678, 2.207)	2.150 (1.740, 2.657)	< 0.001	< 0.001	1.852 (1.413, 2.426)	< 0.001	< 0.001
4-Aminobiphenyl	log (ln)	1.699 (0.215, 2.389)	1.535 (1.294, 1.821)	< 0.001	< 0.001	1.443 (1.152, 1.809)	0.001	0.004
o-Anisidine	log (ln)	-0.551 (-1.127, 0.065)	0.940 (0.809, 1.091)	0.414	0.498	1.043 (0.866, 1.255)	0.660	0.735
2, 6-Dimethylaniline	2 categories	NA	1.402 (1.090, 1.802)	0.008	0.016	1.153 (0.839, 1.583)	0.379	0.533
o-Toluidine	log (ln)	1.312 (0.868, 1.768)	1.196 (1.039, 1.376)	0.012	0.022	1.231 (1.034, 1.464)	0.019	0.047
Arsenics - Speciated								
Arsenous Acid	2 categories	NA	1.107 (0.866, 1.415)	0.416	0.498	1.078 (0.782, 1.488)	0.646	0.735
Arsenobetaine	2 categories	NA	0.924 (0.736, 1.160)	0.495	0.582	0.978 (0.734, 1.305)	0.882	0.920
Arsenocholine	2 categories	NA	0.868 (0.648, 1.162)	0.340	0.428	1.082 (0.748, 1.564)	0.676	0.736
Monomethylarsonic Acid	2 categories	NA	1.070 (0.837, 1.369)	0.589	0.672	0.905 (0.666, 1.230)	0.525	0.661
Dimethylarsinic Acid	log (ln)	-3.303 (-3.760, −2.865)	0.960 (0.833, 1.106)	0.568	0.658	1.142 (0.936, 1.394)	0.191	0.301
Nicotine Metabolites								
Total Cotinine	log (ln)	-5.561 (-6.460, −1.807)	1.859 (1.578, 2.190)	< 0.001	< 0.001	1.849 (1.477, 2.315)	< 0.001	< 0.001
Total Hydroxycotinine	log (ln)	-4.984 (-6.060, −1.273)	1.901 (1.609, 2.247)	< 0.001	< 0.001	1.834 (1.456, 2.310)	< 0.001	< 0.001
TNE-2	log (ln)	-9.703 (-10.714, −6.104)	1.892 (1.600, 2.237)	< 0.001	< 0.001	1.846 (1.466, 2.324)	< 0.001	< 0.001
Ethylene Oxide								
Ethylene Oxide	log (ln)	-1.352 (-2.033, −0.489)	1.486 (1.273, 1.734)	< 0.001	< 0.001	1.521 (1.246, 1.858)	< 0.001	< 0.001
Formaldehyde								
Formaldehyde	log (ln)	0.405 (-0.073, 1.007)	0.778 (0.650, 0.932)	0.006	0.014	0.925 (0.729, 1.175)	0.523	0.661
Heterocyclic Aromatic Amines								
A-α-C	2 categories	NA	1.743 (1.385, 2.194)	< 0.001	< 0.001	1.874 (1.391, 2.524)	< 0.001	< 0.001
Harman	log (ln)	0.468 (-0.136, 1.138)	0.971 (0.833, 1.133)	0.709	0.772	1.160 (0.949, 1.417)	0.146	0.243
MeA-α-C	2 categories	NA	2.857 (2.106, 3.876)	< 0.001	< 0.001	2.643 (1.789, 3.906)	< 0.001	< 0.001
Norharman	log (ln)	5.805 (5.170, 6.482)	1.244 (1.061, 1.460)	0.007	0.015	1.099 (0.899, 1.344)	0.356	0.509
Ph1P	2 categories	NA	1.036 (0.823, 1.305)	0.763	0.819	0.954 (0.712, 1.280)	0.756	0.811
Iodine								
Iodine	log (ln)	0.331 (-0.153, 0.956)	0.984 (0.849, 1.140)	0.826	0.849	1.007 (0.832, 1.220)	0.940	0.953
Metals								
Mercury	log (ln)	-5.627 (-6.338, −4.888)	0.686 (0.580, 0.811)	< 0.001	< 0.001	0.878 (0.706, 1.092)	0.242	0.353
Copper	log (ln)	0.250 (-0.291, 0.845)	0.808 (0.687, 0.951)	0.010	0.019	1.071 (0.861, 1.332)	0.538	0.665
Selenium	log (ln)	0.361 (-0.137, 0.945)	0.815 (0.692, 0.959)	0.013	0.024	1.005 (0.811, 1.244)	0.966	0.966
Zinc	log (ln)	-0.109 (-0.639, 0.464)	0.785 (0.666, 0.926)	0.004	0.009	0.942 (0.760, 1.166)	0.582	0.685
Manganese	2 categories	NA	1.220 (0.937, 1.589)	0.140	0.183	1.104 (0.793, 1.537)	0.558	0.668
Barium	log (ln)	-4.418 (-4.978, −3.782)	0.792 (0.683, 0.917)	0.002	0.004	0.976 (0.802, 1.187)	0.806	0.853
Cadmium	log (ln)	-6.202 (-6.819, −5.660)	1.429 (1.221, 1.672)	< 0.001	< 0.001	1.535 (1.235, 1.909)	< 0.001	< 0.001
Cobalt	log (ln)	-5.496 (-5.893, −5.116)	0.969 (0.847, 1.109)	0.646	0.725	1.265 (1.048, 1.527)	0.013	0.036

(*continued on next page*)

Table 2 (*continued*)

Exposure family by toxicant exposure	Processing	Median (IQR)	Model 1			Model 2		
			OR (95 %CI)	*p-values*	*q-values*	OR (95 %CI)	*p-values*	*q-values*
Cesium	log (ln)	-3.065 (-3.373, −2.730)	0.813 (0.699, 0.947)	0.007	0.015	1.068 (0.859, 1.330)	0.553	0.668
Molybdenum	log (ln)	-0.997 (-1.415, −0.624)	0.989 (0.866, 1.130)	0.871	0.871	1.039 (0.873, 1.237)	0.665	0.735
Lead	log (ln)	-5.621 (-6.093, −5.176)	1.599 (1.371, 1.866)	< 0.001	< 0.001	1.538 (1.254, 1.887)	< 0.001	< 0.001
Antimony	log (ln)	-7.710 (-8.060, −7.292)	1.132 (0.993, 1.290)	0.063	0.092	1.064 (0.902, 1.256)	0.460	0.600
Tin	log (ln)	-5.506 (-6.109, −4.896)	1.119 (0.976, 1.283)	0.105	0.145	0.989 (0.826, 1.183)	0.902	0.927
Strontium	log (ln)	0.017 (-0.430, 0.377)	0.988 (0.865, 1.130)	0.863	0.871	1.177 (0.977, 1.417)	0.085	0.160
Thallium	log (ln)	-6.420 (-6.763, −6.048)	0.735 (0.638, 0.847)	< 0.001	< 0.001	0.954 (0.793, 1.149)	0.621	0.720
Tungsten	log (ln)	-7.531 (-8.103, −6.875)	1.070 (0.920, 1.245)	0.380	0.470	1.076 (0.889, 1.302)	0.450	0.598
Uranium	log (ln)	-9.903 (-10.378, −9.210)	1.132 (0.986, 1.298)	0.077	0.108	1.157 (0.967, 1.383)	0.111	0.188
Polycyclic Aromatic Hydrocarbons								
1-Naphthol	log (ln)	2.677 (2.078, 3.707)	1.450 (1.244, 1.690)	< 0.001	< 0.001	1.418 (1.177, 1.708)	< 0.001	< 0.001
2-Naphthol	log (ln)	3.871 (3.188, 4.570)	1.551 (1.308, 1.838)	< 0.001	< 0.001	1.759 (1.405, 2.203)	< 0.001	< 0.001
3-Hydroxyfluorene	log (ln)	-0.384 (-0.830, 0.476)	1.670 (1.444, 1.930)	< 0.001	< 0.001	1.665 (1.374, 2.019)	< 0.001	< 0.001
2-Hydroxyfluorene	log (ln)	0.499 (0.103, 1.201)	1.639 (1.414, 1.900)	< 0.001	< 0.001	1.690 (1.389, 2.056)	< 0.001	< 0.001
1-Hydroxyphenanthrene	log (ln)	0.100 (-0.327, 0.553)	1.124 (0.975, 1.295)	0.107	0.145	1.230 (1.022, 1.480)	0.027	0.066
1-Hydroxypyrene	log (ln)	0.326 (-0.087, 0.864)	1.313 (1.127, 1.530)	< 0.001	0.001	1.441 (1.174, 1.768)	< 0.001	0.001
2&3-Hydroxyphenanthrene	log (ln)	0.243 (-0.178, 0.705)	1.408 (1.211, 1.636)	< 0.001	< 0.001	1.454 (1.200, 1.762)	< 0.001	< 0.001
Perchlorate, Nitrate & Thiocyanate								
Perchlorate	log (ln)	-3.507 (-3.982, −3.046)	0.847 (0.738, 0.971)	0.016	0.028	0.894 (0.752, 1.063)	0.205	0.305
Nitrate	log (ln)	6.146 (5.768, 6.537)	0.833 (0.729, 0.951)	0.006	0.014	1.077 (0.906, 1.281)	0.401	0.553
Thiocyanate	log (ln)	2.448 (1.826, 3.091)	1.297 (1.128, 1.492)	< 0.001	0.001	1.346 (1.118, 1.621)	0.002	0.005
Volatile Organic Compound Metabolites								
GAMA	2 categories	NA	1.377 (1.089, 1.740)	0.008	0.015	1.273 (0.949, 1.708)	0.107	0.187
HEMA	2 categories	NA	1.451 (1.152, 1.829)	0.002	0.004	1.451 (1.082, 1.945)	0.013	0.036
MHBMA2	2 categories	NA	2.992 (2.106, 4.250)	< 0.001	< 0.001	2.798 (1.799, 4.352)	< 0.001	< 0.001
PHEMA	2 categories	NA	1.671 (1.315, 2.125)	< 0.001	< 0.001	1.367 (1.010, 1.848)	0.043	0.092
PMA	2 categories	NA	1.202 (0.944, 1.530)	0.135	0.180	1.236 (0.915, 1.670)	0.167	0.271
2MHA	log (ln)	-1.003 (-1.780, −0.239)	1.225 (1.035, 1.449)	0.018	0.030	1.192 (0.964, 1.473)	0.104	0.186
3&4 MHA	log (ln)	0.821 (0.061, 1.792)	1.205 (1.009, 1.438)	0.039	0.058	1.098 (0.877, 1.376)	0.413	0.559
AAMA	log (ln)	-0.701 (-1.169, −0.163)	1.284 (1.098, 1.502)	0.002	0.004	1.295 (1.057, 1.587)	0.012	0.036
AMCC	log (ln)	0.515 (-0.109, 1.118)	1.190 (1.022, 1.384)	0.024	0.039	1.245 (1.020, 1.521)	0.031	0.072
ATCA	log (ln)	0.175 (-0.570, 0.760)	0.983 (0.842, 1.147)	0.824	0.849	1.255 (1.009, 1.560)	0.040	0.089
BMA	log (ln)	-2.684 (-3.219, −2.069)	0.843 (0.738, 0.964)	0.012	0.022	0.894 (0.755, 1.059)	0.195	0.301
BPMA	log (ln)	-3.171 (-4.130, −2.336)	0.967 (0.823, 1.137)	0.688	0.761	1.206 (0.981, 1.483)	0.076	0.154
CEMA	log (ln)	-0.101 (-0.592, 0.440)	1.401 (1.219, 1.609)	< 0.001	< 0.001	1.358 (1.140, 1.617)	0.001	0.002
CYMA	log (ln)	-4.013 (-4.547, −2.959)	1.368 (1.229, 1.524)	< 0.001	< 0.001	1.348 (1.170, 1.554)	< 0.001	< 0.001
DHBMA	log (ln)	1.052 (0.760, 1.331)	1.157 (1.024, 1.307)	0.018	0.030	1.172 (1.007, 1.363)	0.038	0.086
2HPMA	log (ln)	-1.296 (-1.793, −0.682)	1.157 (1.010, 1.326)	0.035	0.054	1.311 (1.104, 1.558)	0.002	0.006

(*continued on next page*)

Table 2 (*continued*)

Exposure family by toxicant exposure	Processing	Median (IQR)	Model 1			Model 2		
			OR (95 %CI)	*p-values*	*q-values*	OR (95 %CI)	*p-values*	*q-values*
3HPMA	log (ln)	0.774 (0.285, 1.425)	1.370 (1.180, 1.591)	< 0.001	< 0.001	1.477 (1.218, 1.790)	< 0.001	< 0.001
MA	log (ln)	0.289 (-0.151, 0.707)	1.155 (1.015, 1.314)	0.028	0.044	1.172 (0.995, 1.380)	0.055	0.115
MHBMA3	log (ln)	-2.904 (-3.348, −2.242)	1.483 (1.284, 1.713)	< 0.001	< 0.001	1.523 (1.267, 1.831)	< 0.001	< 0.001
PGA	log (ln)	0.685 (0.328, 1.035)	1.139 (1.007, 1.287)	0.037	0.057	1.106 (0.948, 1.290)	0.198	0.301
HPMMA	log (ln)	0.766 (0.376, 1.548)	1.436 (1.240, 1.663)	< 0.001	< 0.001	1.470 (1.222, 1.769)	< 0.001	< 0.001
TTCA	2 categories	NA	0.804 (0.635, 1.019)	0.071	0.101	0.782 (0.583, 1.049)	0.101	0.183

Model 1 was unadjusted. Model 2 adjusted for age, sex, race/ethnicity, education, marital status, poverty, BMI, alcohol use, dental visit frequency, mouthwash/dental floss use, and eGFR. The *q*-value is a metric for statistical significance that has been corrected for multiple hypothesis testing using the False Discovery Rate (FDR) procedure. It estimates the proportion of false discoveries among all significant findings.

Abbreviations: 1-Naphthol, 1-Hydroxynapthalene; 2&3-Hydroxyphenanthrene, 2-Hydroxyphenanthrene & 3-Hydroxyphenanthrene; 2-Naphthol, 2-Hydroxynapthalene; 2HPMA, N-Acetyl-S- (2-hydroxypropyl)-L-cysteine; 2MHA, 2-Methylhippuric acid; 3&4 MHA, 3- and 4-Methylhippuric acid; 3HPMA, N-Acetyl-S- (3-hydroxypropyl)-L-cysteine; A-α-C, 2-Amino-9H-pyrido[2,3-*b*]indole; AAMA, N-Acetyl-S- (2-carbamoylethyl)-L-cysteine; AMCC, N-Acetyl-S- (N-methylcarbamoyl)-L-cysteine; ATCA, 2-Aminothiazoline-4-carboxylic acid; BMA, N-Acetyl-S- (benzyl)-L-cysteine; BPMA, N-Acetyl-S- (n-propyl)-L-cysteine; CEMA, N-Acetyl-S- (2-carboxyethyl)-L-cysteine; CYMA, N-Acetyl-S- (2-cyanoethyl)-L-cysteine; DHBMA, N-Acetyl-S- (3,4-dihydroxybutyl)-L-cysteine; ExWAS, exposome-wide association study; GAMA, N-Acetyl-S- (2-carbamoyl-2-hydroxyethyl)-L-cysteine; HEMA, N-Acetyl-S- (2-hydroxyethyl)-L-cysteine; HPMMA, N-Acetyl-S- (3-hydroxypropyl-1-methyl)-L-cysteine; MA, Mandelic acid; MHBMA2, N-Acetyl-S- (2-hydroxy-3-butenyl)-L-cysteine; MHBMA3, N-Acetyl-S- (4-hydroxy-2-butenyl)-L-cysteine; MeA-α-C, 2-Amino-3-methyl-9H-pyriodo[2,3-*b*]indole; PGA, Phenylglyoxylic acid; PHEMA, N-Acetyl-S- (phenyl-2-hydroxyethyl)-L-cysteine.

influenced by unobserved confounders, such as unmeasured lifestyle factors, environmental exposures, or genetic predispositions. Using residual correlation coefficients (ρ) to quantify the impact of potential confounding, we found that as the sensitivity parameter ρ increased—reflecting greater potential confounding—the average causal mediation effects (ACME) for all biomarkers, while varying in magnitude, remained statistically significant across a wide range of ρ values (Supplementary Figure 4).

Furthermore, we examined how the mediation effects might be influenced by varying levels of R^2_M (coefficient of determination for the mediator model) and R^2_Y (coefficient of determination for the outcome model), which represent the proportion of variance explained by unobserved confounders. The results suggested that the mediation effects remained relatively robust under these sensitivity scenarios (Supplementary Figure 4.). Finally, although mediation analyses provide valuable mechanistic insights, they cannot establish definitive causal pathways, thereby requiring further validation through experimental studies.

3.7. Sensitivity and robustness analyses

To strengthen analytical robustness, sensitivity ExWAS analyses were performed using unimputed datasets (Supplementary Figure 5 and Supplementary Table 7).

Outlier assessment confirmed that the majority of pollutants exhibited acceptable outlier proportions (<10 %), and the results were not driven by influential observations (Supplementary Table 1).

To assess the potential confounding effect of tobacco smoking, we performed two sets of analyses. First, we observed extensive and significant correlations between smoking status and multiple environmental pollutants (Supplementary Figure 6). Against this background, the associations of most pollutants were substantially attenuated after additional adjustment for smoking status. Conversely, analysis restricted to the subpopulation of never-smokers identified lead, 2HPMA, and 2-Naphthol as being significantly and positively associated with periodontitis risk (Supplementary Table 8–9).

The robustness of the identified toxicant mixture was mutually reinforced by the WQS and ERS approaches. The WQS model identified 2-Naphthol (weight = 0.098), Lead (weight = 0.078), and 2-Aminonaphthalene (weight = 0.030) as the primary contributors to the overall mixture effect. Notably, these high-weight compounds in the WQS analysis also exhibited significant non-zero coefficients in the ERS construction, demonstrating a high degree of methodological concordance. Furthermore, several additional compounds—including MeA-α-C, MHBMA2, TTCA, Manganese, A-α-C, and Thiocyanate—were consistently identified as important mixture components by both analytical methods, reinforcing the stability of these specific toxicants within the complex exposure profile (Supplementary Figure 7).

4. Discussion

In this first exposome-wide association study of periodontitis among U.S. adults, we systematically evaluated the relationship between complex environmental mixtures and periodontal disease. Our analysis identified 29 environmental toxicants across nine chemical categories significantly associated with periodontitis risk after rigorous false discovery rate correction. Key individual toxicants included MHBMA2 (OR = 2.798), 2-Aminonaphthalene (OR = 1.852 per IQR), and Total Cotinine (OR = 1.849 per IQR). Furthermore, an environmental risk score (ERS) constructed from 17 priority toxicants demonstrated a robust dose-response relationship with periodontitis, with each IQR increase in ERS associated with 59.7 % higher disease risk (OR = 1.597) and individuals in the high ERS group facing a 2.552-fold increased risk. Stratified analyses revealed heightened susceptibility among males and adults under 60 years of age. Mediation analyses indicated that systemic inflammatory biomarkers explained 5.05 %–11.49 % of the association between ERS and periodontitis. These findings underscore the role of environmental toxicant mixtures as potentially modifiable risk factors for periodontal disease.

The observed associations between environmental toxicants and periodontitis align with established mechanistic pathways involving oxidative stress and inflammatory dysregulation. Acrylamide, a toxic substance widely found in industrial products and the environment, is primarily metabolized through two pathways in the body: one is conjugation with glutathione (GSH) to form non-toxic metabolites; the other is bioactivation by cytochrome P450 enzymes (CYP2E1) to form the more toxic glycidamide(Cantrell and McDougal, 2021). Our results indicated a significant positive correlation between acrylamide exposure and the incidence of periodontitis. Studies have demonstrated that glycidamide can induce excessive reactive oxygen species (ROS)

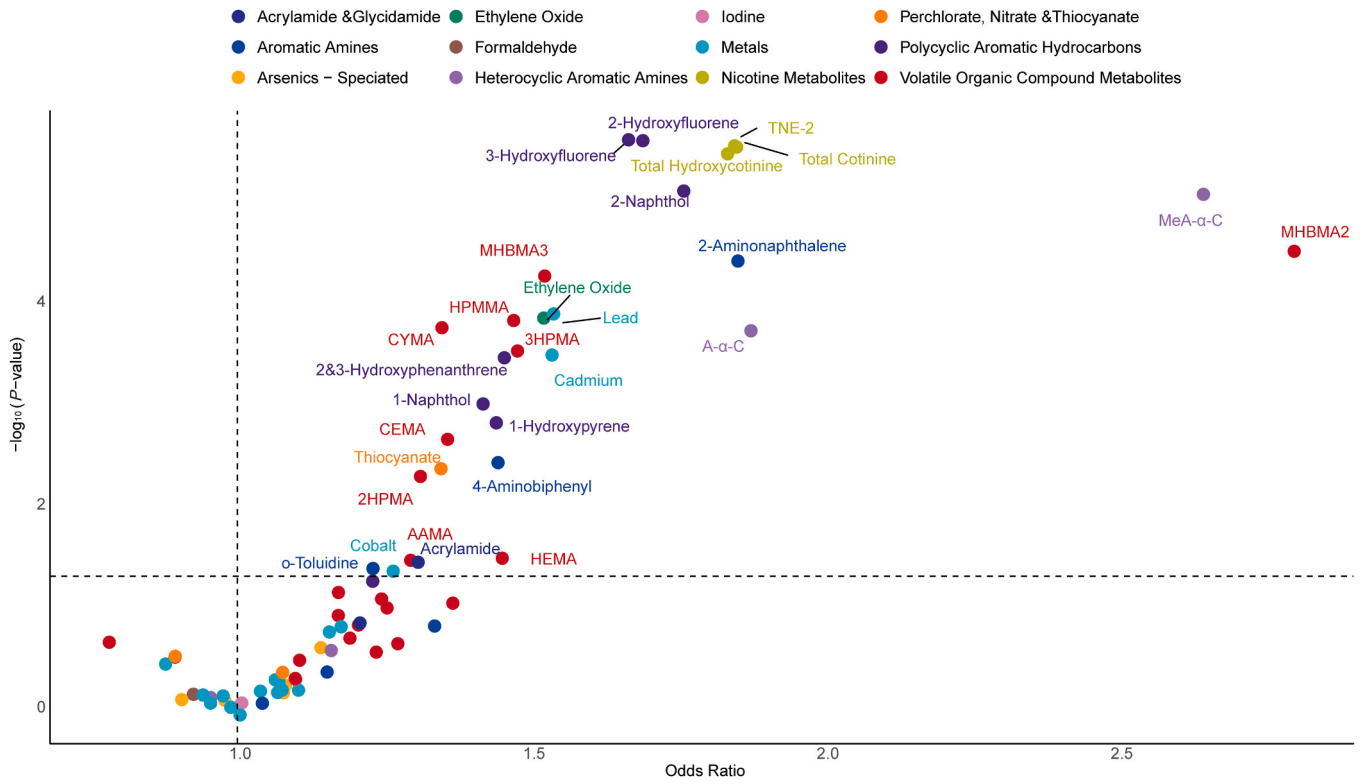


Fig. 2. Exposome-wide association study of 73 environmental toxicants exposures with periodontitis among 1210 United States adults. Note: Adjusted for age, sex, race/ethnicity, education, marital status, poverty, BMI, alcohol use, dental visit frequency, mouthwash/dental floss use, and eGFR. Measurement units by chemical category are as follows: Acrylamide & Glycidamide: pmol/g Hb; Aromatic Amines: pg/mL; Arsenic Species: μg/L; Nicotine Metabolites: Total Cotinine and Total Hydroxycotinine in ng/mL, TNE-2 in nmol/mL; Ethylene Oxide: pmol/g Hb; Formaldehyde: nmol/g Hb; Heterocyclic Aromatic Amines: pg/mL; Iodine: ng/mL; Metals: all metals in μg/L except Mercury in ng/mL; Polycyclic Aromatic Hydrocarbons: ng/L; Perchlorate, Nitrate & Thiocyanate: ng/mL; Volatile Organic Compound Metabolites: μg/L. Abbreviations: 1-Naphthol, 1-Hydroxynapthalene; 2&3-Hydroxyphenanthrene, 2-Hydroxyphenanthrene & 3-Hydroxyphenanthrene; 2-Naphthol, 2-Hydroxynapthalene; 2HPMA, N-Acetyl-S- (2-hydroxypropyl)-L-cysteine; 2MHA, 2-Methylhippuric acid; 3&4 MHA, 3- and 4-Methylhippuric acid; 3HPMA, N-Acetyl-S- (3-hydroxypropyl)-L-cysteine; A-α-C, 2-Amino-9H-pyrido[2,3-*b*]indole; AAMA, N-Acetyl-S- (2-carbamoylethyl)-L-cysteine; AMCC, N-Acetyl-S-(N-methylcarbamoyl)-L-cysteine; ATCA, 2-Aminothiazoline-4-carboxylic acid; BMA, N-Acetyl-S- (benzyl)-L-cysteine; BMI, body mass index; BPMA, N-Acetyl-S- (n-propyl)-L-cysteine; CEMA, N-Acetyl-S- (2-carboxyethyl)-L-cysteine; CYMA, N-Acetyl-S- (2-cyanoethyl)-L-cysteine; DHBMA, N-Acetyl-S- (3,4-dihydroxybutyl)-L-cysteine; eGFR, estimated glomerular filtration rate; GAMA, N-Acetyl-S- (2-carbamoyl-2-hydroxyethyl)-L-cysteine; HEMA, N-Acetyl-S- (2-hydroxyethyl)-L-cysteine; HPMMA, N-Acetyl-S- (3-hydroxypropyl-1-methyl)-L-cysteine; MA, Mandelic acid; MHBMA2, N-Acetyl-S- (2-hydroxy-3-butenyl)-L-cysteine; MHBMA3, N-Acetyl-S- (4-hydroxy-2-butenyl)-L-cysteine; MeA-α-C, 2-Amino-3-methyl-9H-pyriodo[2,3-*b*]indole; PGA, Phenylglyoxylic acid; PHEMA, N-Acetyl-S- (phenyl-2-hydroxyethyl)-L-cysteine.

generation within cells, triggering oxidative stress responses that lead to DNA damage, lipid peroxidation, and apoptosis(Juan et al., 2021). The core pathological feature of periodontitis is the destruction of periodontal supporting tissues caused by dysregulated host immune-inflammatory responses to plaque biofilms, which may indirectly explain the observed correlation(Sczepanik et al., 2020).

Similarly, ethylene oxide (EO)-a prevalent toxicant-undergoes hydrolytic metabolism to form ethylene glycol(Moayed and Mahdavian, 2017). It also conjugates with glutathione (GSH), generating S-carboxymethylglutathione and S-carboxymethylhomocysteine. This depletes GSH reserves, induces oxidative stress, and may ultimately contribute to periodontitis pathogenesis(Miao et al., 2024). Aromatic amine toxicants may be converted via multiple metabolic pathways in the body into substances with oxidant and pro-inflammatory properties, thereby damaging periodontal tissues(Bellamri et al., 2021). The development of periodontitis may be promoted by cellular oxidative stress induced by oxidative metabolites of polycyclic aromatic hydrocarbons(Shamsedini et al., 2022). Furthermore, in vitro evidence indicates that PAH exposure participates in the pathophysiological processes of periodontitis (Monnouchi et al., 2016). Volatile Organic Compounds demonstrate high volatility, environmental mobility, and degradation resistance (David and Niculescu, 2021). Human exposure pathways include direct oral contact via inhalation or periodontal tissue interaction through ingestion of contaminated food/water(Kwon et al., 2018). In vitro studies establish that aromatic VOC exposure triggers inflammatory responses through NF-κB activation and reactive oxygen species-mediated oxidative stress(Bönisch et al., 2012).

A recent scoping review emphasized that periodontal tissue damage from heavy metals (e.g., lead and cadmium) involves oxidative stress, inflammatory responses, and dysregulated bone metabolism-consistent with this meta-analysis(Coelho et al., 2025). Corroborating this, Huang et al. (2022) reported elevated periodontitis prevalence in a large U.S. cohort with increased blood levels of trace elements containing heavy metals, reinforcing environmental pollutants' role in periodontal disease(Huang et al., 2022). Nicotine, a toxic constituent of tobacco, represents a significant risk factor for periodontitis(Isik Andrikopoulos et al., 2019). Experimental evidence indicates that nicotine markedly enhances autophagic activity in human periodontal ligament cells (hPDLCs) and promotes IL-1β and IL-8 secretion through α7 nicotinic acetylcholine receptor (nAChR) signaling, thereby exacerbating inflammatory responses in periodontal tissues and advancing periodontitis pathogenesis(Du et al., 2021). While this biological plausibility supports the observed associations, a methodological caution is warranted in their interpretation. Specifically, when nicotine metabolites (Total Cotinine, Total Hydroxycotinine, TNE-2) and other environmental exposures are simultaneously included as predictor variables in regression

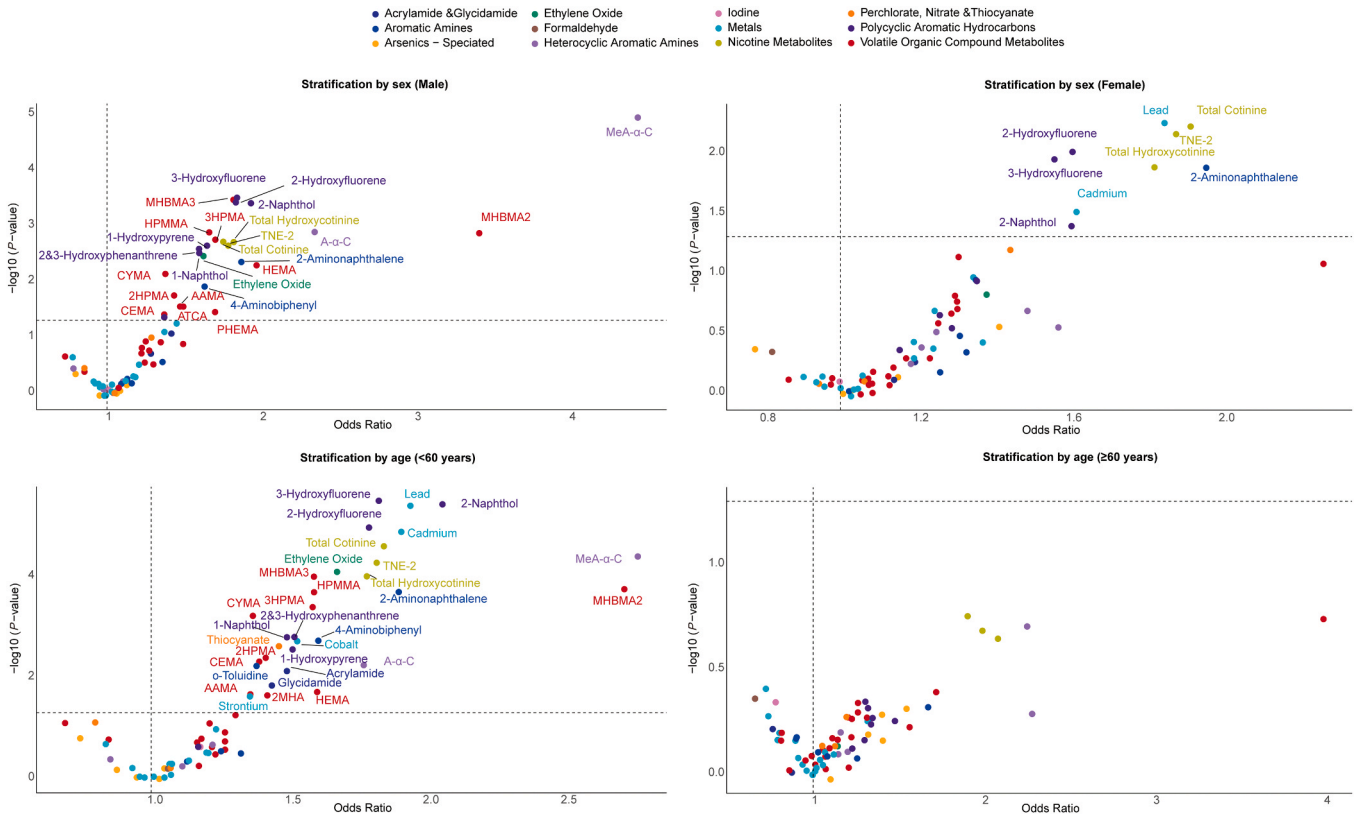


Fig. 3. Adjusted association between environmental toxicants (73 exposures) and periodontitis in 1210 adults (exposome-wide association study analysis) stratified by age, and sex. Note: Adjusted for age, sex, race/ethnicity, education, marital status, poverty, BMI, alcohol use, dental visit frequency, mouthwash/dental floss use, and eGFR. Measurement units by chemical category are as follows: Acrylamide & Glycidamide: pmol/g Hb; Aromatic Amines: pg/mL; Arsenic Species: μg/L; Nicotine Metabolites: Total Cotinine and Total Hydroxycotinine in ng/mL, TNE-2 in nmol/mL; Ethylene Oxide: pmol/g Hb; Formaldehyde: nmol/g Hb; Heterocyclic Aromatic Amines: pg/mL; Iodine: ng/mL; Metals: all metals in μg/L except Mercury in ng/mL; Polycyclic Aromatic Hydrocarbons: ng/L; Perchlorate, Nitrate & Thiocyanate: ng/mL; Volatile Organic Compound Metabolites: μg/L. Abbreviations: 1-Naphthol, 1-Hydroxynapthalene; 2&3-Hydroxyphenanthrene, 2-Hydroxyphenanthrene & 3-Hydroxyphenanthrene; 2-Naphthol, 2-Hydroxynapthalene; 2HPMA, N-Acetyl-S- (2-hydroxypropyl)-L-cysteine; 2MHA, 2-Methylhippuric acid; 3&4 MHA, 3- and 4-Methylhippuric acid; 3HPMA, N-Acetyl-S- (3-hydroxypropyl)-L-cysteine; A-α-C, 2-Amino-9H-pyrido[2,3-*b*]indole; AAMA, N-Acetyl-S- (2-carbamoylethyl)-L-cysteine; AMCC, N-Acetyl-S- (N-methylcarbamoyl)-L-cysteine; ATCA, 2-Aminothiazoline-4-carboxylic acid; BMA, N-Acetyl-S- (benzyl)-L-cysteine; BMI, body mass index; BPMA, N-Acetyl-S- (n-propyl)-L-cysteine; CEMA, N-Acetyl-S- (2-carboxyethyl)-L-cysteine; CYMA, N-Acetyl-S- (2-cyanoethyl)-L-cysteine; DHBMA, N-Acetyl-S- (3,4-dihydroxybutyl)-L-cysteine; eGFR, estimated glomerular filtration rate; GAMA, N-Acetyl-S- (2-carbamoyl-2-hydroxyethyl)-L-cysteine; HEMA, N-Acetyl-S- (2-hydroxyethyl)-L-cysteine; HPMMA, N-Acetyl-S- (3-hydroxypropyl-1-methyl)-L-cysteine; MA, Mandelic acid; MHBMA2, N-Acetyl-S- (2-hydroxy-3-butenyl)-L-cysteine; MHBMA3, N-Acetyl-S- (4-hydroxy-2-butenyl)-L-cysteine; MeA-α-C, 2-Amino-3-methyl-9H-pyriodo[2,3-*b*]indole; PGA, Phenylglyoxylic acid; PHEMA, N-Acetyl-S- (phenyl-2-hydroxyethyl)-L-cysteine.

models, conditioning on these variables—which share "smoking" as a common cause—may induce collider bias. This statistical phenomenon can open non-causal pathways between exposures or distort the true effect estimates. Therefore, although mechanistically supported, the independent associations for nicotine metabolites should be interpreted with this limitation in mind. Overall, sufficient evidence supports our research into the exposome of toxicants. Implementing targeted measures based on these findings may play a crucial role in the prevention and treatment of periodontitis.

The differential susceptibility patterns observed across demographic groups warrant particular attention. Males and younger adults demonstrated greater numbers of significant toxicant-periodontitis associations compared with females and older individuals. This pattern may reflect sex-specific differences in toxicant metabolism and detoxification capacity, as estrogen activates the Nrf2 pathway, enhancing antioxidant enzyme expression and reducing female susceptibility to oxidative toxicants(Straub, 2007). Additionally, younger adults may experience higher acute exposure levels due to occupational and lifestyle factors, emphasizing the importance of targeted surveillance in this population (Chu et al., 2025).

Mediation analyses revealed that environmental toxicants may influence periodontitis through immunomodulatory mechanisms. Specifically the association between toxicant exposure and periodontitis is partially mediated by white blood cell count, neutrophil count, monocytes, and composite inflammatory indices (SII, NLR, SIRI). These factors function in the body's immune and inflammatory responses. Studies demonstrate that inflammation stimulates immune cells to produce chemokines and cytokines, which recruit monocytes, leukocytes, and polymorphonuclear neutrophils to sites of oxidative stress (Mörbt et al., 2009). While the mediation proportions in this study were relatively modest (5 %-11 %), they provide mechanistic insights into potential pathways linking environmental exposures to periodontal disease.

To our knowledge, this represents the first ExWAS examining comprehensive environmental toxicant exposure in relation to periodontitis, analyzing 73 toxicants across 12 chemical categories using nationally representative data. However, this study has several limitations. First, the cross-sectional design of NHANES precludes causal inference and cannot rule out reverse causality. The use of a single survey cycle (2013–2014) also limits the assessment of long-term exposure trends and within-individual variation. Second, for most toxicants, exposure assessment relied on single spot urine measurements, which may lead to exposure misclassification, particularly for compounds with short half-lives, as a single measurement may not

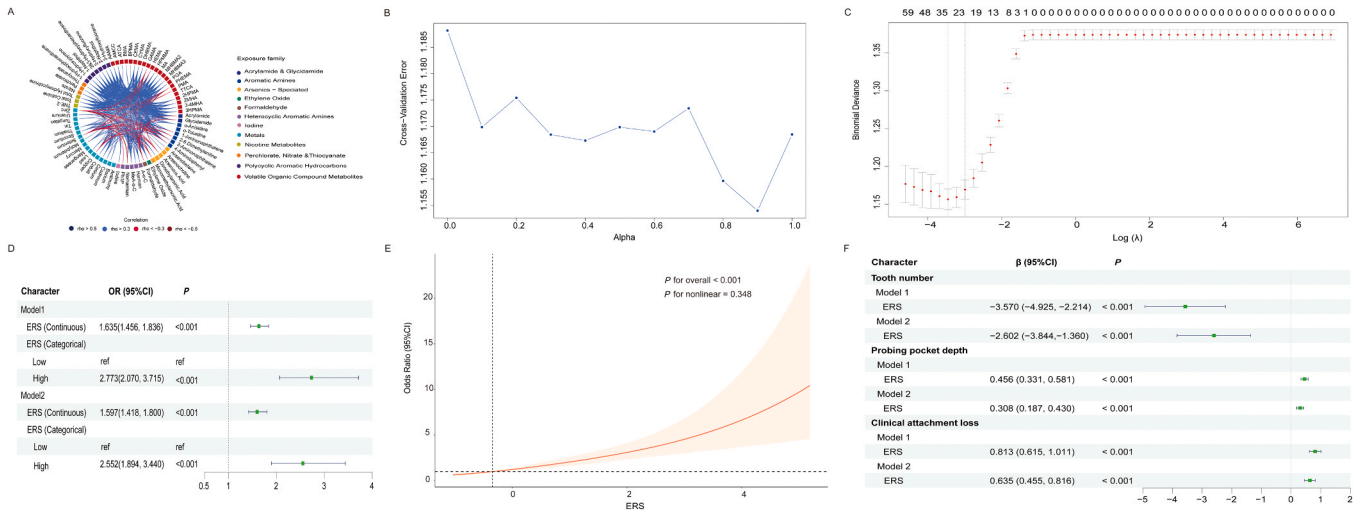


Fig. 4. (A) Correlation of 73 environmental toxicants exposures; (B) Alpha tuning in elastic net regression via cross-validation; (C) Lambda selection in elastic net: binomial deviance across Log(λ). 17 environmental toxicants (2-Aminonaphthalene, 2-Naphthol, 3-Hydroxyfluorene, A-α-C, BMA, CYMA, Iodine, Lead, Manganese, MeA-α-C, MHBMA2, o-Toluidine, Thiocyanate, TNE-2, Total Cotinine, and TTCA) were selected; (D) Forest plot illustrating the logistic regression analysis of the association between the environmental risk score and periodontitis. (E) Restricted cubic spline plots of the environmental risk score and periodontitis. (F) Forest plot illustrating the linear regression analysis of the association between the environmental risk score and tooth number, probing pocket depth, and clinical attachment loss. Model 1 was unadjusted. Model 2 adjusted for age, sex, race/ethnicity, education, marital status, poverty, BMI, alcohol use, dental visit frequency, mouthwash/dental floss use, and eGFR. Abbreviation: ERS, environmental risk score. Abbreviations: 2-Aminonaphthalene, 2-Naphthol, 2-Hydroxynapthalene; 3-Hydroxyfluorene, A-α-C, 2-Amino-9H-pyrido[2,3-*b*]indole; BMA, N-Acetyl-S- (benzyl)-L-cysteine; BMI, body mass index; CYMA, N-Acetyl-S- (2-cyanoethyl)-L-cysteine; eGFR, estimated glomerular filtration rate; MeA-α-C, 2-Amino-3-methyl-9H-pyriodo[2,3-*b*]indole; MHBMA2, N-Acetyl-S- (2-hydroxy-3-butenyl)-L-cysteine.

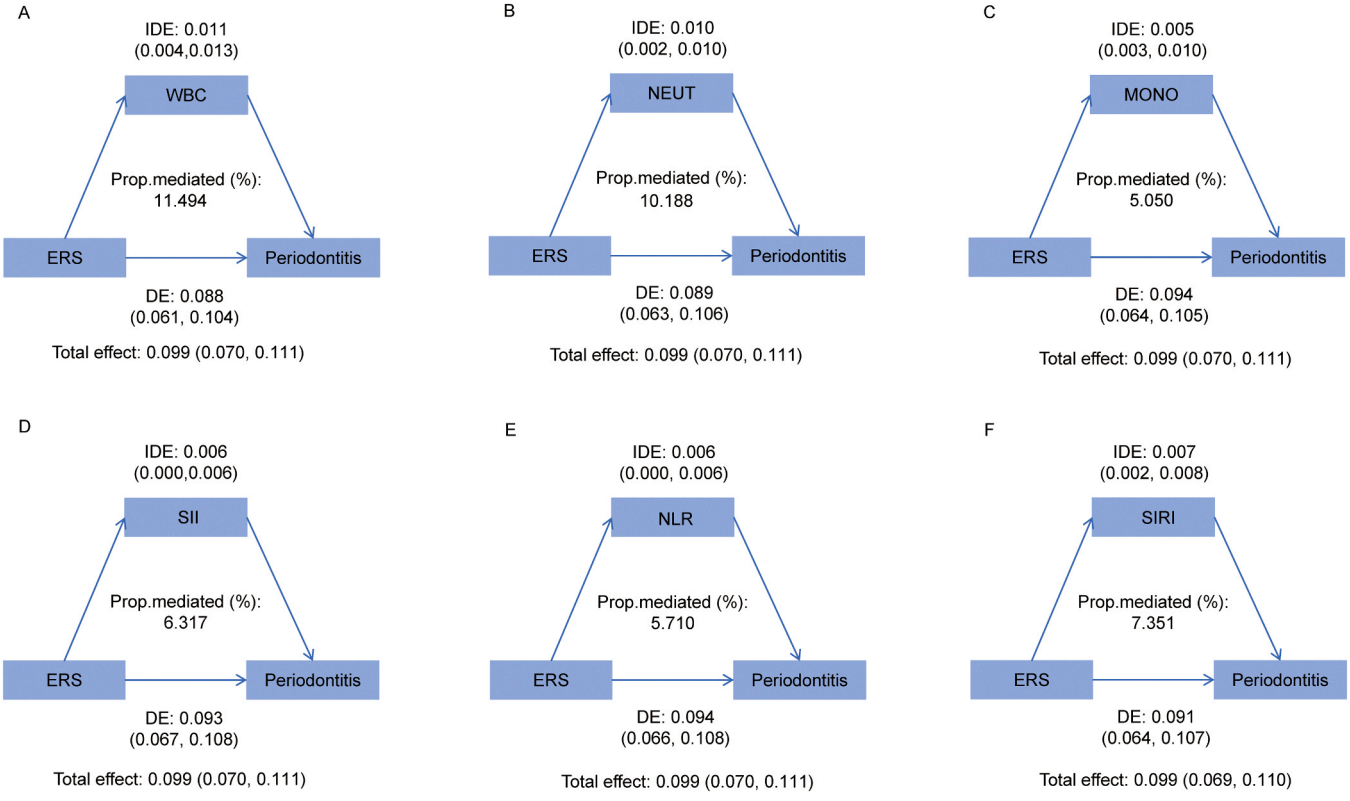


Fig. 5. The mediating roles of systemic inflammatory biomarkers in the association between the environmental risk score and periodontitis. (A) For white blood cell counts; (B) For neutrophils; (C) For monocytes. (D) For systemic immune-inflammation index; (E) For neutrophil-to-lymphocyte ratio; (F) For systemic inflammatory response index. Abbreviation: DE, direct effect; ERS, environmental risk score; IDE, indirect effect; MONO, monocyte; NEUT, neutrophil; NLR, neutrophil-to-lymphocyte ratio; SII, systemic immune-inflammation index; SIRI, systemic inflammatory response index; WBC, white blood cell.

accurately reflect long-term internal exposure levels. Third, the 73 toxicants investigated represent only a fraction of the human exposome; future studies would benefit from broader suspect screening or non-targeted analytical techniques. Fourth, although NHANES data are nationally representative, the generalizability of our findings to non-U.S. populations may be limited due to differences in environmental regulations, lifestyle factors, and genetic backgrounds. Finally, despite extensive confounder adjustment, residual confounding cannot be entirely ruled out, although E-values suggest that unmeasured confounding is unlikely to fully explain the observed associations (Supplementary Table 11). The mediation analyses provide preliminary mechanistic insights but require further validation through longitudinal or experimental studies.

5. Conclusions

This exposome-wide association study identified 29 environmental toxicants across 9 chemical categories that were significantly associated with periodontitis risk after false discovery rate correction. Notable individual toxicants included the MHBMA2 (OR = 2.798) and 2-Aminonaphthalene (OR = 1.852 per IQR). The constructed environmental risk score (ERS), comprising 17 priority toxicants, demonstrated a robust dose-response relationship with periodontitis, where each IQR increase was associated with a 59.7 % higher disease risk (OR = 1.597). Stratified analyses revealed heightened susceptibility among males and adults younger than 60 years. Exploratory mediation analysis suggested that systemic inflammatory biomarkers partially mediate these associations. These findings highlight environmental toxicant exposure as a potentially modifiable risk factor for periodontitis, informing both public health policy and clinical prevention strategies. Future research should focus on prospective validation of these associations and elucidation of underlying biological mechanisms through experimental approaches.

CRediT authorship contribution statement

Xuejing Zhong: Writing – review & editing, Writing – original draft, Conceptualization. **Xixi Dong:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Fa Chen:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Lingjun Yan:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Yanfen Li:** Writing – review & editing, Validation, Supervision. **Fuhua Yan:** Writing – review & editing, Validation, Supervision. **Jing Wang:** Writing – review & editing, Validation, Supervision. **Baochang He:** Writing – review & editing, Validation, Supervision. **Yichen Lin:** Writing – review & editing, Validation, Supervision. **Jinmei Wu:** Writing – review & editing, Writing – original draft, Conceptualization.

Ethics approval and consent to participate

The National Center for Health Statistics Research Ethics Review Board reviewed and approved NHANES, and all participants provided written informed consent.

Consent for publication

No applicable.

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Declaration of Competing Interest

The authors declare that they have no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2025.119544.

Data Availability

Data from the National Health and Nutrition Examination Survey (NHANES) 2013–2014 are publicly available online (https://www.cdc.gov/nchs/nhanes/ index.htm).

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